Evaluating Large Language Models for Clinical Text Neutralisation: A Comparative Study of Medical LLMs in Spanish-to-English

Abstract

This report evaluates large language models (LLMs) for neutralising Spanish clinical exam justifications into coherent clinical narratives. Three models (MedLlama-2-7B, Qwen2.5-1.5B, and Mistral-7B-Instruct) were tested across six prompt designs using a structured scoring framework. Mistral-7B emerged as the most reliable (mean score 3.96/6), demonstrating superior fidelity, neutralisation quality, and clinical clarity. The study identifies critical failure patterns in medical fact preservation, particularly hallucination of clinical terms, fabricated findings, and temporal distortions, and proposes future work on hallucination detection and cross-lingual embedding improvements.

1. Introduction

1.1 Context

Large language models have demonstrated remarkable text generation capabilities, yet their application to specialised medical text processing in low-resource languages remains underexplored. Clinical exam justifications contain option-based elimination reasoning that must be converted into coherent clinical narratives. This neutralisation task requires preserving medical facts while removing exam-style formatting, testing both linguistic and domain-specific LLM capabilities.

This study emerges from work examining LLM performance on Irish-language clinical text, where initial observations revealed significant challenges in

language support. Castilho et al.'s taxonomy for analysing hallucinations in low-resource language outputs provided a systematic evaluation framework, classifying invented words into distinct patterns: verb hallucinations (with or without Irish morphological compliance), and six noun hallucination types including compounds, lazy Gaelicisation (phonetic translations using Irish orthography), good confabulations (morphologically correct inventions where no official translation exists), code-switching (English words with Irish suffixes), and prefix/suffix-based creations. This taxonomy distinguished between confabulations (following Irish morphological rules) and violations of those rules, enabling systematic identification of hallucination patterns.

1.2 Research Objectives

This internship aimed to:

1. Develop a reproducible evaluation framework for clinical text neutralisation quality assessment
2. Compare performance across medical-specialist, generalist, and small-scale LLMs
3. Identify systematic failure patterns in medical fact preservation
4. Establish baseline metrics for future model improvement and hallucination detection

2. Technical Set-up & Constraints

2.1 Infrastructure Evolution

Initial experiments utilised local inference via Ollama on consumer hardware. On Windows, only TinyLlama could be successfully run before switching the laptop's OS to Fedora Linux, which enabled testing of additional models including Llama2-7B, Llama3.1-8B, Gemma3, and MedLlama2-7B. Local infrastructure proved insufficient for two reasons: (1) hardware limitations prevented loading models larger than 8B parameters, and (2) multilingual models failed to generate usable Irish-language output despite various prompting strategies.

2.2 Cloud Migration

Google Colab's free tier (≈15GB GPU memory) enabled testing larger models but still imposed constraints. Attempting to evaluate UCCIX-13B (Irish LLM) resulted in consistent out-of-memory errors, which shifted the focus on models within the 1.5B to 7B parameter range.

2.3 Final Model Selection

Three models were selected representing different capability tiers:

- **MedLlama-2-7B**: Medical domain specialist
- **Mistral-7B-Instruct**: Competent generalist with strong instruction-following
- **Qwen2.5-1.5B**: Stress-test baseline for small model behaviour

This gradient enables analysis of how model size, domain specialisation, and general capability affect clinical neutralisation quality.

3. Methodology

3.1 Evaluation Framework

A three-dimensional scoring system assessed clinical text neutralisation:

3.1.1 Fidelity (0 to 2 points)

Preservation of all medical facts, evidence markers, and logical links; absence of hallucinations or fabricated clinical details; retention of temporal qualifiers, lab values, and diagnostic criteria.

3.1.2 Neutralisation Quality (0 to 2 points)

Complete removal of option numbers and exam-style references; elimination of contrastive "incorrect option" reasoning; production of unified, option-free clinical narrative.

3.1.3 Clinical Clarity & Coherence (0 to 2 points)

Natural clinical reasoning flow; appropriate medical terminology and logic; absence of contradictions or structural breaks.

Maximum composite score: 6 points

3.2 Prompt Engineering Strategy

Six prompt architectures tested different instruction-following paradigms:

Prompt 1: System + One-Shot Demonstration - Persona-based prompt with single example. Testing revealed high sensitivity to example content, with smaller models copying demonstrations verbatim.

Prompt 2: Zero-Shot Instruction - Pure instruction-based approach with explicit preservation rules. Successfully reduced hallucinations through "no new information" constraints.

Prompt 3: Delimiter-Based - High-directive format using ### markers. Reduced drift and improved focus on rewriting versus invention.

Prompt 4: Do/Don't Constraint - Explicit permitted/forbidden behaviour lists. Clear rules prevented drift but required careful constraint specification.

Prompt 5: Anti-CoT Final-Answer - Requested direct output without reasoning steps. Produced superior style but catastrophic fidelity failures as models invented clinical content.

Prompt 6: Output-Targeted - Specified exact output structure (5 to 8 sentence narrative). Achieved grammatical clarity but struggled with fact preservation.

3.3 Test Cases

Test cases were randomly sampled from the CC-MIR Spanish medical exam dataset using `random.sample(data, 2)`, selecting two cases per evaluation run. Across all test iterations, twelve clinical specialisations emerged: Cardiology, Paediatrics, Digestive Surgery, Neurology, Endocrinology, Gynaecology, Nephrology, Emergency Medicine, Psychiatry, Infectious Disease, Plastic Surgery, and Anatomy. Each case required neutralising option-based exam justifications into coherent clinical explanations whilst preserving all medical facts.

4. Results

4.1 Overall Performance

Mean composite scores across all prompts and cases:

- **MedLlama-2-7B**: 2.58/6 (43%)
- **Qwen2.5-1.5B**: 2.58/6 (43%)
- **Mistral-7B**: 3.96/6 (66%)

Mistral-7B demonstrated a performance advantage over both competitors, establishing it as the most reliable model for clinical neutralisation tasks (see Appendix A, Figure 4).

4.2 Dimensional Performance Analysis

4.2.1 Fidelity Scores

- **MedLlama**: 0.67/2 (33%) - Frequent hallucinations and factual distortions
- **Qwen**: 0.71/2 (36%) - Occasional correct outputs amid severe failures
- **Mistral**: 1.25/2 (63%) - Most consistent fact preservation

4.2.2 Neutralisation Quality

- **MedLlama**: 1.00/2 (50%) - Successfully removed options but retained structure
- **Qwen**: 1.12/2 (56%) - Best neutralisation performance
- **Mistral**: 1.34/2 (67%) - Consistent option-free narratives

4.2.3 Clinical Clarity

- **MedLlama**: 0.92/2 (46%) - Inconsistent coherence
- **Qwen**: 0.75/2 (38%) - Frequent logical breaks
- **Mistral**: 1.37/2 (69%) - Superior clinical reasoning flow

Dimensional comparisons appear in Appendix A, Figures 3-4 and 8.

4.3 Case-Specific Performance

Perfect scores (6/6): Turner Syndrome (Prompt 3, Qwen), Cardio B (Prompt 2, Qwen), Paliativos B (Prompt 2, multiple models)

Complete failures (0 to 1/6): Gine 2 (Prompt 5, Qwen: 0/6), Pediatría (Prompt 1, Qwen: 0/6), Hematología (Prompt 1, Qwen: 1/6), multiple cases with Anti-CoT

prompt design

Performance by case and specialisation visualised in Appendix A, Figures 1-2.

4.4 Prompt Performance Analysis

Best performing: Zero-Shot Instruction (Prompt 2) - Mean score 4.5/6; explicit preservation rules reduced hallucinations; clear neutralisation instructions improved output structure.

Worst performing: System + One-Shot (Prompt 1) - Mean score 1.0/6; models copied example content verbatim; long examples overwhelmed smaller models.

Detailed prompt-by-prompt performance for MedLlama appears in Appendix A, Figures 5 and 12.

4.5 Key Failure Patterns

4.5.1 Hallucination Types

Invented clinical terms: Models generated non-existent medical vocabulary (e.g., "daño ateroespérmico", "carbonic acid lithium", "intravenous lithium chelation therapy")

Fabricated lab values and findings: Models added clinical findings absent from source text (e.g., hypokalemia/hyperkalemia, hemopneumothorax instead of pleural effusion)

Temporal distortions: Models altered timing qualifiers (e.g., "within hours" became "asymptomatic for several days", acute/chronic reversals)

Anatomical misrepresentation: Models confused anatomical structures or omitted critical details (e.g., oesophageal vs local sclerosis, arterial collateral pathways)

4.5.2 Neutralisation Failures

Despite instructions, models frequently retained "Option 1", "Option 2" references, maintained eliminative reasoning ("incorrect because..."), and produced paragraph-by-paragraph option analysis rather than unified narratives.

4.6 MedLlama English-Language Performance

To investigate whether MedLlama-2-7B's underwhelming Spanish performance resulted from language mismatch, additional testing used English medical case data (the model was trained primarily on English biomedical corpora).

4.6.1 Overall English Performance

English testing averaged **3.51/6** (58.5%), representing **36% improvement** over Spanish (2.58/6). Across quality dimensions:

- **Clarity**: 1.23/2 (61.5%), highest dimension
- **Fidelity**: 1.15/2 (57.5%)
- **Neutralisation**: 1.13/2 (56.5%)

This improvement confirms language-dependent performance variance, but MedLlama's English scores still fell short of Mistral-7B's Spanish performance (3.96/6), suggesting general instruction-following capability remains more critical than language alignment alone (see Appendix A, Figures 7-8).

4.6.2 Specialisation-Dependent Performance

Performance varied dramatically by clinical domain (see Appendix A, Figure 11):

- **Cardiology**: ≈5.5/6; **Traumatology**: ≈5.5/6 (strongest)
- **Pneumology**: ≈4/6; **Digestive**: ≈4/6 (mid-range)
- **Neurology**: ≈2/6; **Nephrology**: ≈2/6
- **Emergencies**: ≈0.7/6 (poorest)

This suggests MedLlama's reasoning capabilities are highly domain-dependent, possibly reflecting uneven speciality representation in training corpora or fundamental differences in reasoning complexity across domains.

4.6.3 Cross-Language Comparison

English substantially outperformed Spanish across all dimensions (see Appendix A, Figure 7):

- **Fidelity**: 2.5× improvement (1.15/2 vs 0.67/2)
- **Clarity**: 2.5× improvement (1.23/2 vs 0.92/2)
- **Neutralisation**: 1.2× improvement (1.13/2 vs 1.00/2)

The substantially larger Fidelity and Clarity improvements suggest MedLlama's medical knowledge representations are strongly anchored to English linguistic patterns. While the model can perform structural transformations (neutralisation) across languages, preserving medical facts and maintaining clinical coherence requires operating within its primary training distribution.

4.6.4 Computational Resource Comparison

English showed more stable resource utilisation (see Appendix A, Figures 9-10):

Processing time per prompt:

- **English**: 2 to 11 minutes (P1: 11 min → P6: 2 min, generally decreasing)
- **Spanish**: More variable patterns

RAM consumption:

- **English**: 7.0 to 7.3 GB (consistent)
- **Spanish**: Greater variability

Both languages peaked at Prompt 3, indicating this case-prompt combination was computationally intensive regardless of language.

4.6.5 Implications

These findings confirm language mismatch contributes significantly to MedLlama's performance limitations but does not fully explain them. Even with English input, MedLlama underperforms Mistral-7B processing Spanish text, reinforcing that **instruction-following capability and general linguistic flexibility outweigh domain-specific pretraining** for clinical text transformation tasks.

4.7 Resource Utilisation Summary

Processing time per prompt (mean, Spanish evaluation):

- MedLlama: 18 to 20 minutes
- Qwen: 1 to 3.5 minutes
- Mistral: 10 to 12 minutes

RAM consumption (Spanish evaluation):

- MedLlama: ~7 GB (consistently high)

- Qwen: 5 to 6 GB (variable)
- Mistral: 6 to 7 GB (stable)

Efficiency analysis: Mistral offers optimal performance-resource balance (see Appendix A, Figures 6 and 16).

5. Discussion

5.1 What Worked

5.1.1 Model Selection

Mistral-7B's superior performance validates that general instruction-following capability matters more than medical domain pretraining for structured rewriting tasks. MedLlama's medical specialisation did not compensate for weaker instruction adherence.

5.1.2 Prompt Design Insights

Zero-shot instruction prompts with explicit constraint lists outperformed demonstration-based approaches. Key success factors included clear "preserve all facts" rules that reduced omissions, "no new information" constraints that suppressed hallucinations, and framing the task as "rewrite" rather than "answer" to prevent inferential reasoning. Delimiter-based prompts (###) reduced model drift by creating structural boundaries.

5.1.3 Evaluation Framework

The three-dimensional scoring system successfully captured distinct failure modes. Models could achieve high neutralisation while failing fidelity (removing options but inventing facts), or high clarity masking fidelity problems (fluent but incorrect narratives). Composite scoring revealed no model excelled across all dimensions simultaneously.

5.2 What Didn't Work

5.2.1 Medical Domain Specialisation

MedLlama-2-7B's medical training did not translate to better performance due to two critical mismatches:

Language Distribution Mismatch - MedLlama trained on large-scale biomedical text (PubMed abstracts, PubMed Central articles, medical textbooks, clinical guidelines) overwhelmingly in English. While this training shifts internal representations toward medical terminology and clinical discourse patterns, it creates a fundamental disadvantage when processing Spanish clinical text. The model's medical knowledge representations are anchored to English linguistic patterns, forcing operation outside its primary training distribution.

Genre Distribution Mismatch - MedLlama's training focused on academic and biomedical literature, not exam-style clinical reasoning. Research papers and clinical guidelines differ fundamentally from multiple-choice exam justifications: biomedical literature follows research conventions, structured documentation, and hierarchical knowledge presentation, whereas exam justifications employ contrastive eliminative reasoning ("Option A is incorrect because..."), comparative diagnosis analysis, and pedagogical explanation style.

Implications - MedLlama's training optimised it for understanding medical concepts in English academic contexts. The neutralisation task required transforming exam-style reasoning into narrative form, following structured text rewriting instructions, and processing Spanish input. This explains why Mistral-7B (a multilingual generalist with superior Spanish capabilities and instruction-following capacity) outperformed MedLlama despite lacking medical specialisation.

5.2.2 Prompt Strategies

One-shot demonstration prompts failed catastrophically with smaller models, producing verbatim example copies rather than applying demonstrated patterns. Anti-CoT prompts (requesting output without reasoning) achieved excellent style but terrible fidelity: models filled information gaps with plausible-sounding fabrications.

5.2.3 Model Scaling

Qwen2.5-1.5B's small parameter count proved insufficient for reliable clinical reasoning. While achieving occasional perfect scores, performance was highly unstable across cases.

5.3 Systematic Challenges

5.3.1 The Fidelity-Neutralisation Trade-off

Models face competing pressures: preserve all medical facts intact versus remove option-based structure and create narrative flow. When forced to choose, models defaulted to fluency over fidelity, producing coherent but medically incorrect text.

5.3.2 Hallucination Mechanisms

Two distinct patterns emerged. Type 1 (gap-filling hallucinations): when restructuring text, models invented bridging content to improve narrative flow (e.g., adding "haemodynamic monitoring" absent from source). Type 2 (confident confabulation): models generated medically formatted but entirely fabricated content with appropriate terminology, making errors difficult to detect without source comparison.

5.4 Implications for Clinical NLP

This work demonstrates that medical-specific models are not automatically superior for structured text tasks. Instruction-following capability is critical for controlled generation. Hallucination risk increases when models must simultaneously transform and preserve content. Evaluation requires multi-dimensional frameworks capturing distinct failure modes.

6. Conclusion & Future Work

6.1 Key Findings

1. **Mistral-7B outperforms both medical-specialist and small generalist models** for clinical text neutralisation, achieving 66% accuracy versus 43% for competitors
2. **Fidelity preservation is the primary failure mode** across all models (mean 33 to 36%), despite adequate neutralisation performance (50 to 56%)
3. **Prompt design significantly impacts performance**: zero-shot instruction prompts outperformed demonstration-based approaches by 3.5 points (4.5/6 vs 1.0/6)

4. **Small models (≤2B parameters) are unreliable** for clinical reasoning despite occasional perfect scores
5. **Medical domain specialisation does not guarantee superior performance** for text transformation tasks requiring instruction-following

6.2 Limitations

Only three models were tested due to hardware limitations; evaluation covered Spanish-to-English only; the sample comprised 12 cases across 6 prompts (72 total evaluations per model); a single evaluator conducted all assessments; limited optimisation cycles were possible due to time constraints.

6.3 Future Work

6.3.1 Hardware Constraints as Research Advantage

Rather than viewing limited hardware as a mere constraint, this research demonstrates that small models serve as valuable diagnostic tools for understanding clinical LLM failures. The 7B parameter models tested here exhibit hallucinations and fidelity failures more clearly than larger models, where superior capabilities may mask underlying reasoning weaknesses. If verification mechanisms and prompt strategies successfully mitigate failures at this scale, solutions will generalise to larger models while remaining computationally accessible for clinical deployment in resource-constrained settings.

6.3.2 Irish Medical LLM Development

Building on work by McInerney (2025) and Tran et al. (2024) on Irish language models, future work should explore:

Medical Domain Fine-tuning - Adapt existing Irish LLMs (Qomhrá, UCCIX) for clinical text processing; evaluate baseline performance on Irish medical translation and neutralisation tasks; document Irish-language clinical reasoning failure patterns.

Synthetic Data Creation Strategies - Develop multilingual medical datasets combining Irish, English, and Spanish clinical content; follow established Irish LLM methodologies with domain-specific medical corpora curation; explore back-translation and parallel corpus generation.

Data Annotation Recommendations - Design medical terminology crash courses for annotators; create annotation guidelines preserving clinical accuracy; establish quality control protocols for medical translation validation.

6.3.3 Logical Verification Layer Tool

A promising direction for hallucination prevention involves creating a lightweight verification system treating clinical reasoning as formal logic:

Architecture:

1. **Predicate Logic Extraction**: Convert natural language clinical statements into logical atoms (Diagnosis(patient, condition), Symptom(patient, finding), Test(patient, result), Treatment(patient, intervention))
2. **Inference Validation Engine**: Verify every Diagnosis(p, d) is supported by at least one Symptom(p, s) or Test(p, t); confirm every Treatment(p, tx) is supported by at least one Diagnosis(p, d); flag unsupported inferences
3. **Contradiction Detection**: Apply rule-based contradictions (e.g., "acute" vs "chronic" for same condition); use medical ontologies (SNOMED-CT, ICD) to detect incompatible terms; verify temporal consistency

Implementation Benefits: Post-processing layer; model-agnostic; computationally lightweight; provides interpretable failure explanations; iteratively improvable with domain-specific rules. This addresses core fidelity failures identified in this study.

6.3.4 Hallucination Detection Tool

A Python-based automated hallucination detection system is proposed:

Phase 1: Term Extraction - Extract all medical terms, lab values, and clinical entities from source and output; identify terms present in output but absent from source; flag terms with modified qualifiers (acute→chronic).

Phase 2: Classification - Categorise hallucinations by type (invented term, modified value, temporal distortion, anatomical error); calculate hallucination frequency and severity metrics; build hallucination glossary for model retraining.

Phase 3: Embedding Improvement - Use hallucination corpus to identify cross-lingual embedding weaknesses; retrain embeddings with corrective examples; test whether improved embeddings reduce hallucination rates.

6.3.5 Expanded Model Comparison

Future work should evaluate larger models (13B to 70B parameters), recent medical-specialist models (BioGPT, Med-PaLM variants), instruction-tuned MedLlama versions, and non-English base models for multilingual capabilities.

6.3.6 Enhanced Evaluation Framework

Automated metrics development: BERT-score for semantic preservation, medical entity alignment scoring, structural neutralisation metrics.

Expanded test set: 100+ cases across 20+ medical specialities, multiple difficulty tiers, cross-language validation.

6.3.7 MedLlama Re-evaluation

Given MedLlama was evaluated first, potential evaluator bias should be addressed by re-scoring MedLlama outputs alongside other models, conducting blind evaluation across all models, and including additional evaluators.

6.4 Broader Impact

This work contributes to clinical NLP methodology through reproducible evaluation frameworks for medical text transformation; advances low-resource language processing by providing insights into model behaviour under linguistic constraints; enriches LLM evaluation science with multi-dimensional scoring systems capturing distinct failure modes; and supports medical AI safety through documentation of hallucination patterns in clinical contexts.

References

Castilho, S., Fitzsimmons, Z., Holton, C., & Mc Donagh, A. (2025). Synthetic Fluency: Hallucinations, Confabulations, and the Creation of Irish Words in LLM-Generated Translations. *LREC-COLING 2025*.

McInerney, J. (2025). Qomhrá: A Bilingual Irish-English Large Language Model. *arXiv preprint arXiv:2510.17652*.

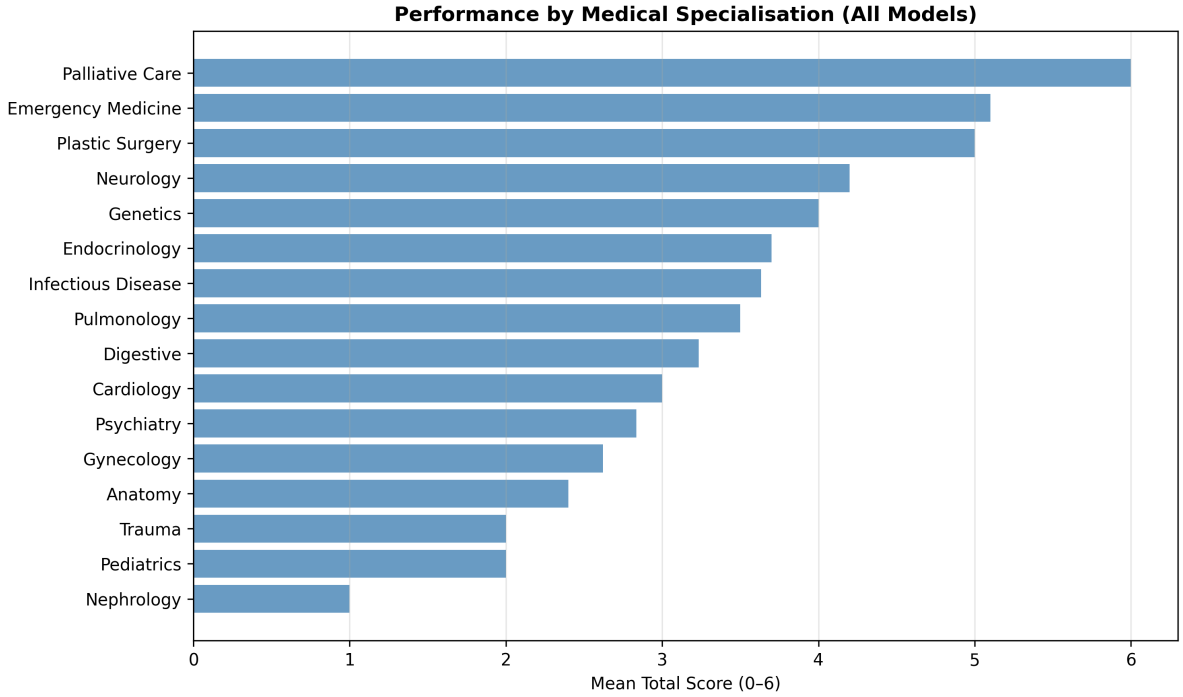
Tran, K.-T., O'Sullivan, B., & Nguyen, H. D. (2024). UCCIX: Irish-eXcellence Large Language Model. *arXiv preprint arXiv:2405.13010*.

Appendices

Appendix A: Performance Visualisations

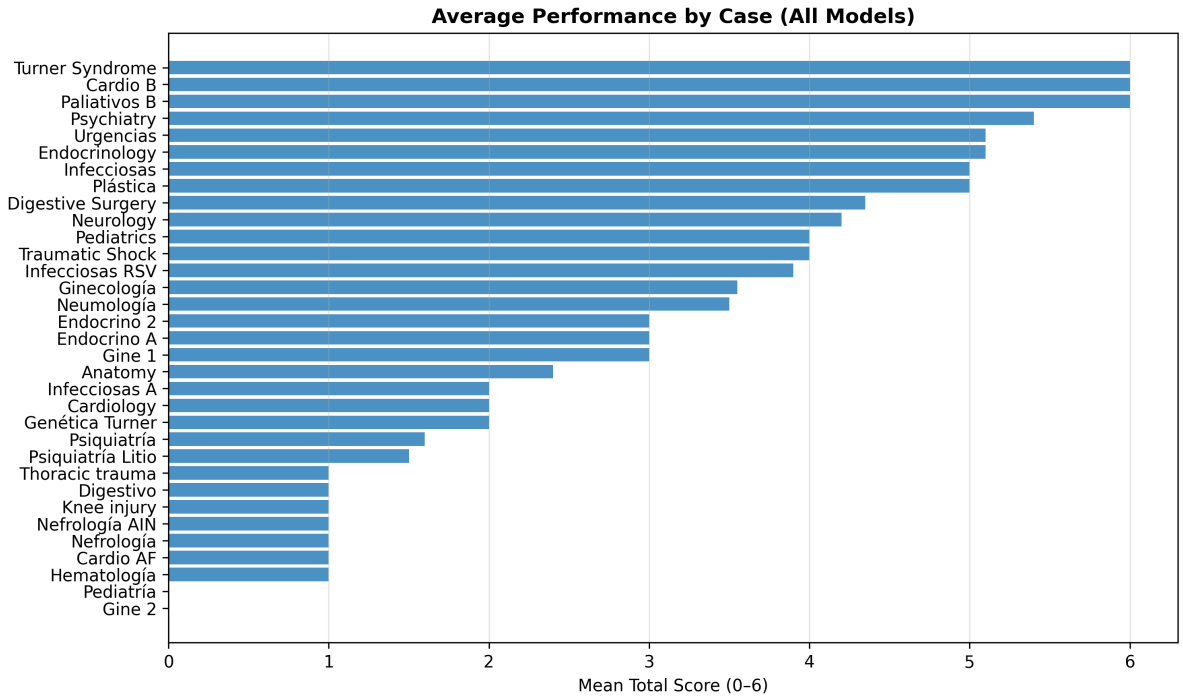
A.1 Spanish Evaluation Results

Figure 1: Performance by Medical Specialisation (All Models)



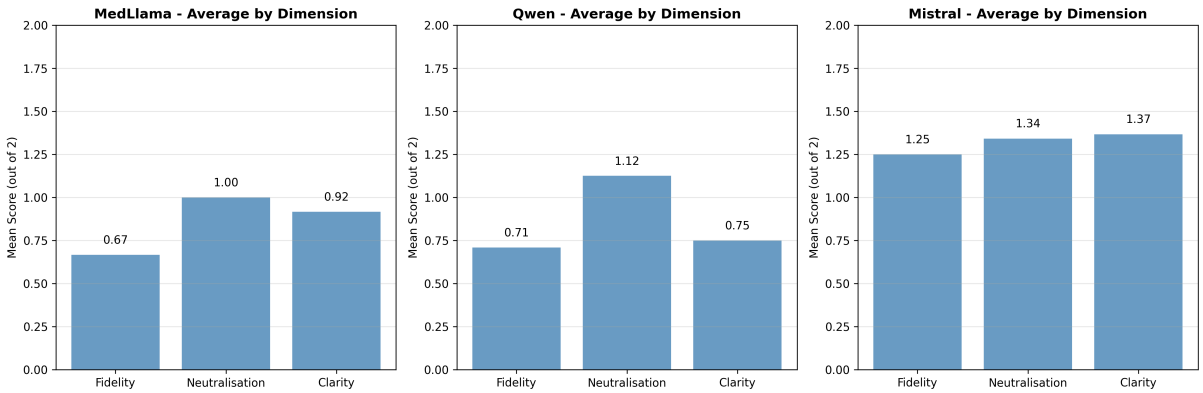
Performance across medical specialities showing Palliative Care and Emergency Medicine achieved highest scores (≈6/6) while Nephrology had lowest performance (≈1/6)

Figure 2: Average Performance by Clinical Case



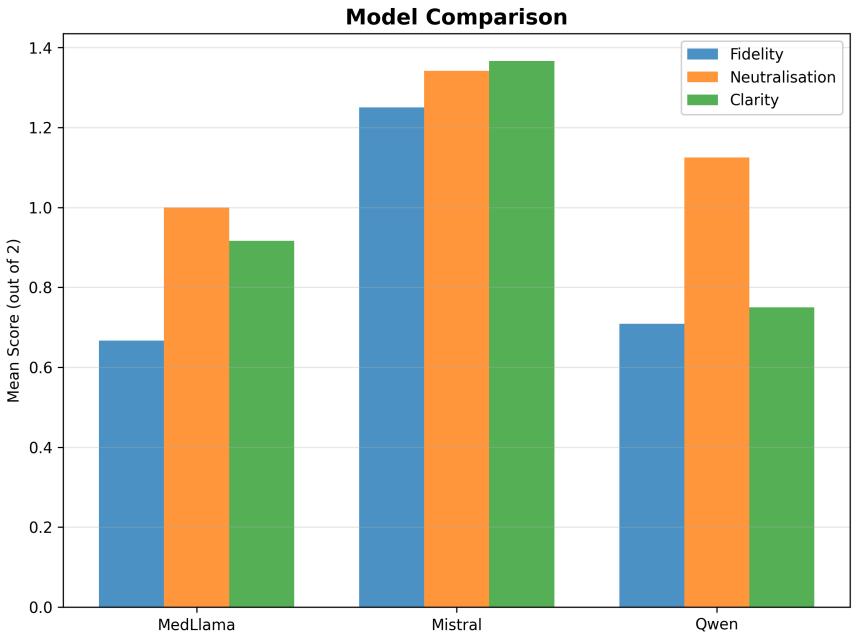
Turner Syndrome, Cardio B, and Paliativos B achieved perfect or near-perfect scores while Gine 2, Pediatría cases showed critical failures

Figure 3: Dimensional Score Comparison Across Models



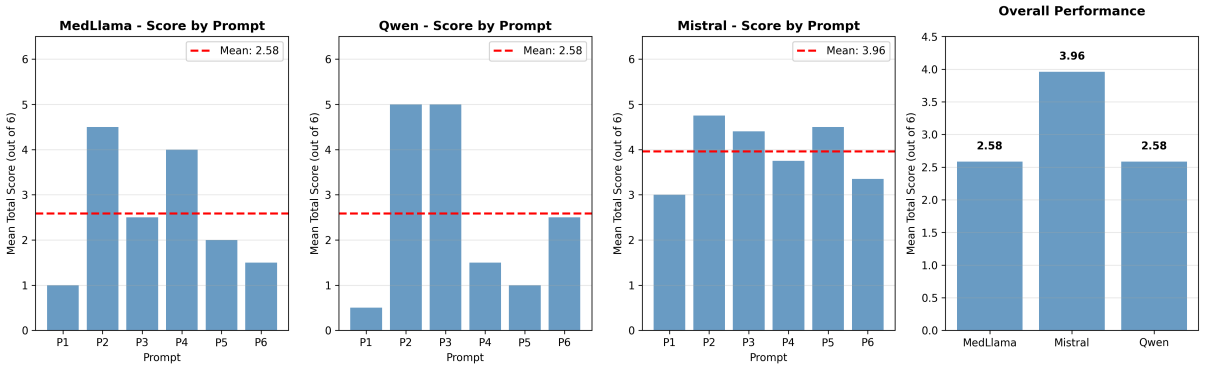
Three-panel comparison showing Mistral leads across all dimensions (Fidelity: 1.25, Neutralisation: 1.34, Clarity: 1.37) compared to MedLlama and Qwen

Figure 4: Overall Model Performance Summary



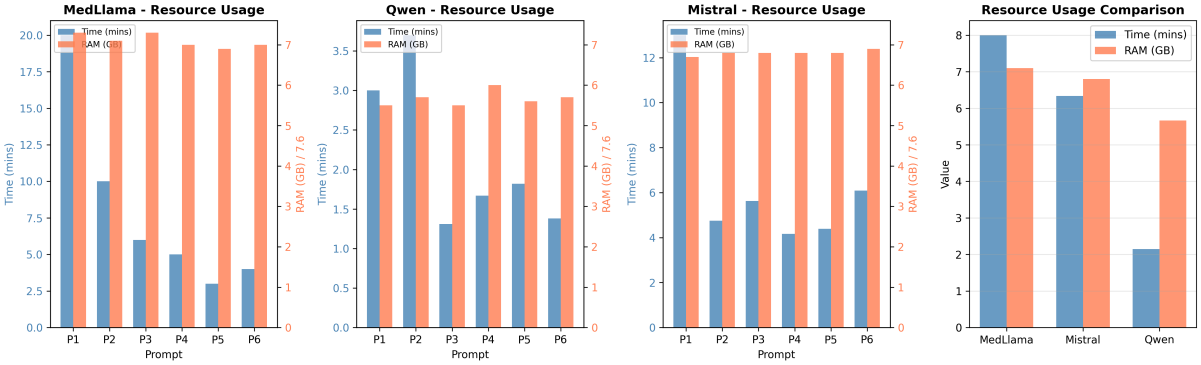
Bar chart showing Mistral's superior performance across Fidelity (1.25), Neutralisation (1.34), and Clarity (1.37) dimensions

Figure 5: MedLlama Performance by Prompt Design



Four-panel figure showing prompt-specific performance with Prompt 2 (Zero-Shot) achieving highest mean scores (4.5/6) and overall performance summary

Figure 6: Computational Resource Usage Comparison



Four-panel analysis of time and RAM consumption per prompt showing MedLlama most resource-intensive, Qwen fastest but inconsistent, Mistral balanced

A.2 Cross-Language Performance Analysis

Figure 7: English vs Spanish Performance by Dimension (MedLlama)

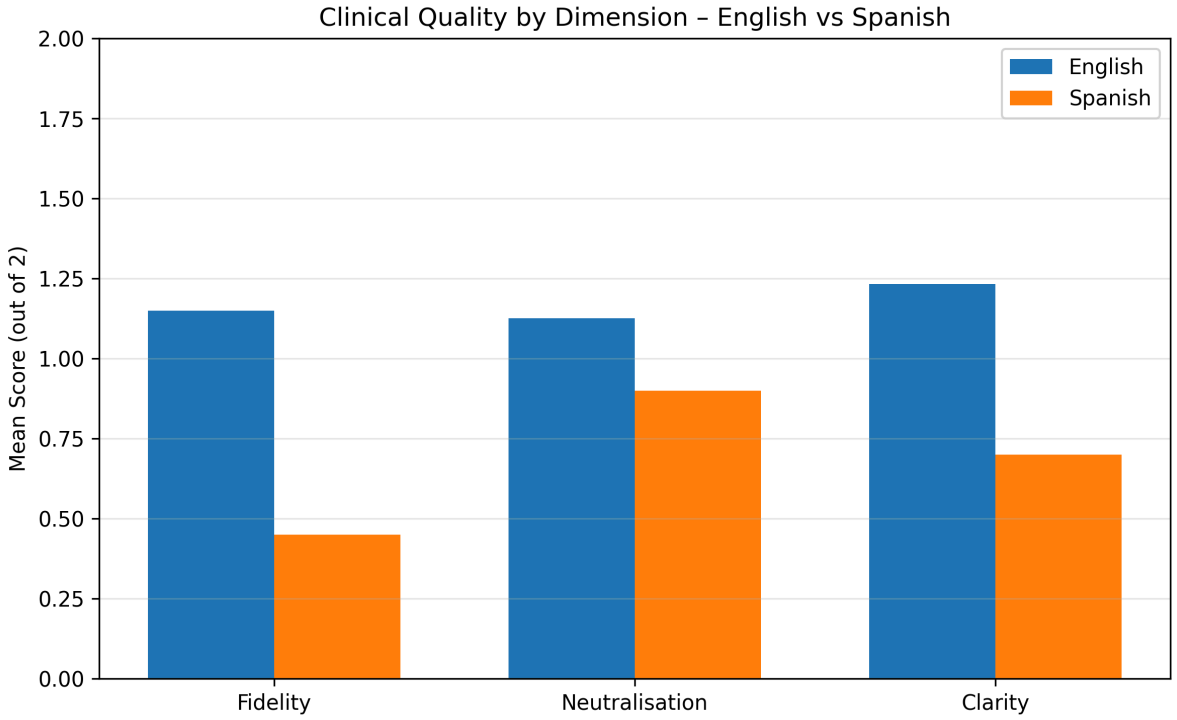
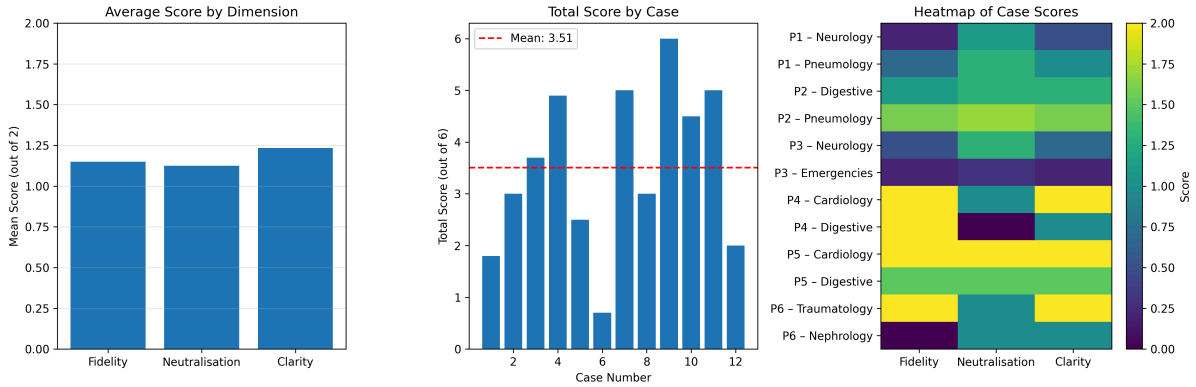


Figure 8: MedLlama English Clinical Quality Overview



A.3 Computational Resource Analysis

Figure 9: RAM Usage Comparison (English vs Spanish)

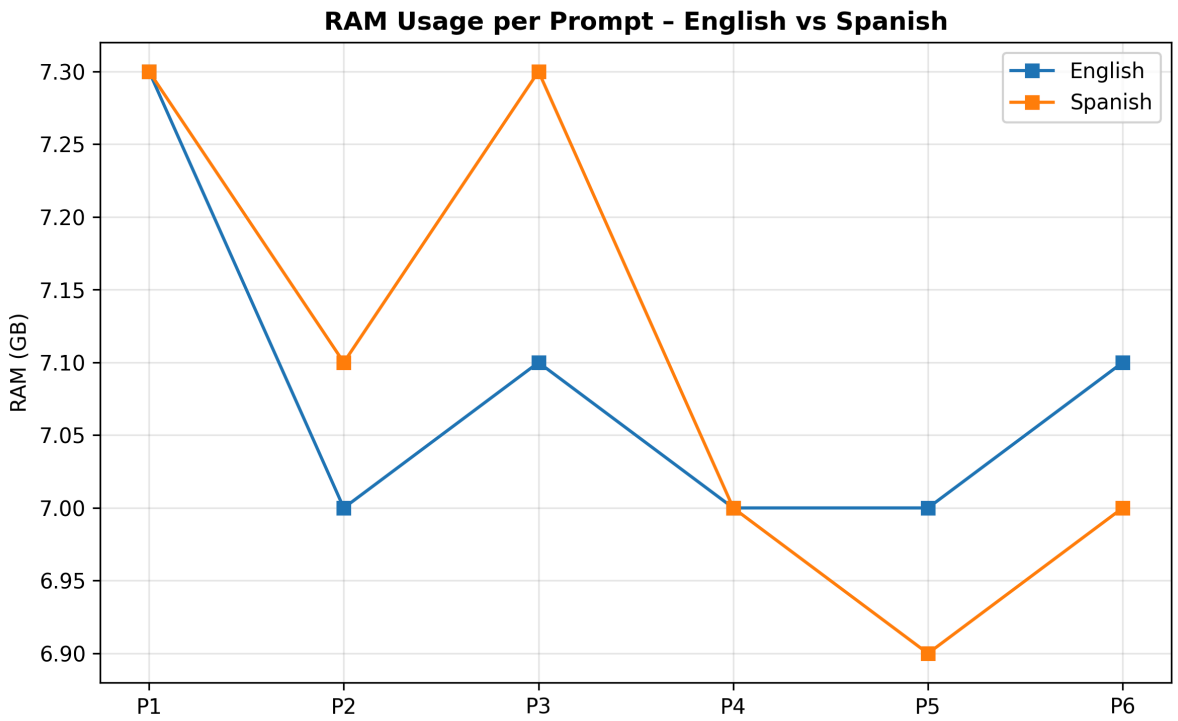
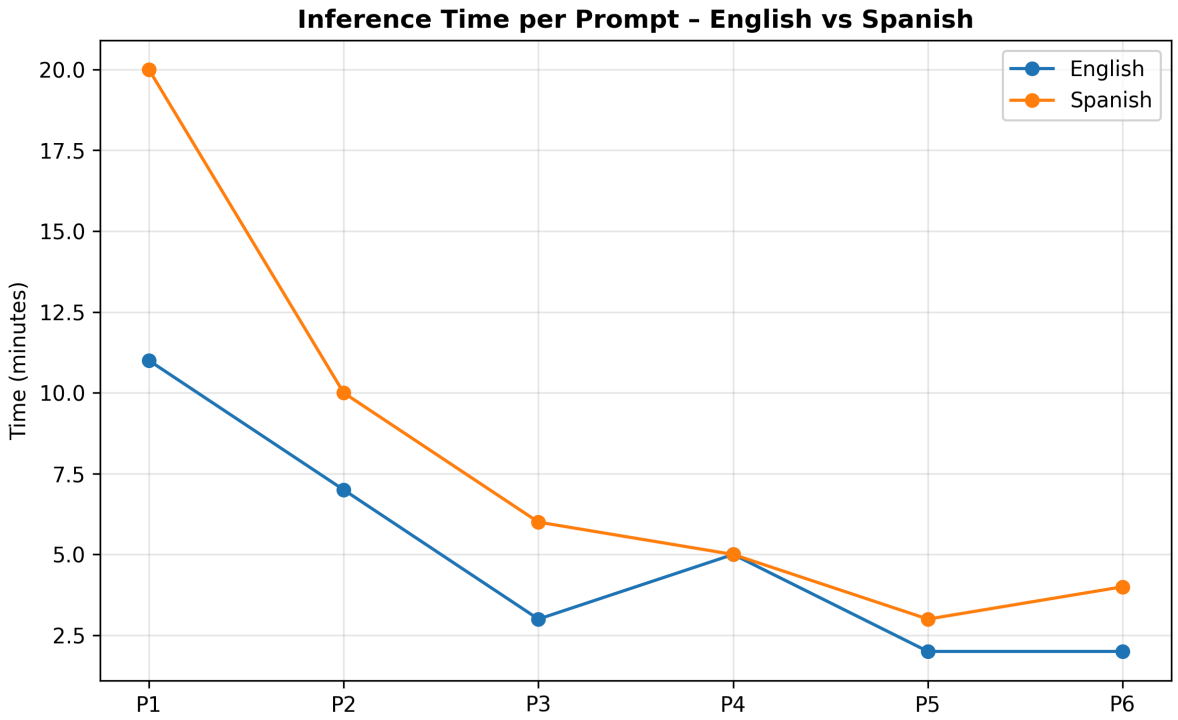
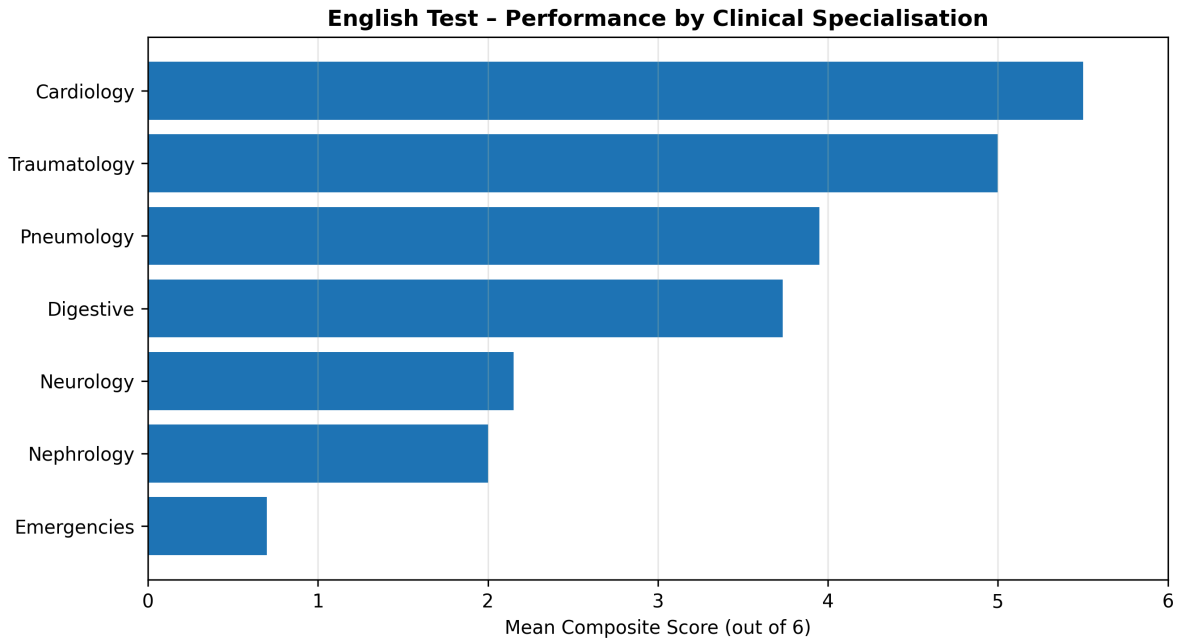


Figure 10: Inference Time Comparison (English vs Spanish)



A.4 Domain-Specific Performance

Figure 11: Performance by Medical Specialisation (MedLlama English)



Appendix B: Detailed Failure Examples

Example 1: Severe Hallucination (Qwen, Nephrology Case)

Source: Drug-induced renal injury with arthralgias, oliguria, proteinuria 1.2 g/day

Output: "daño ateroespérmico" with "necrosis tubular por trombos presenta un mecanismo inmunológicamente inducido"

Issues: Invented non-existent term, fabricated mechanisms, omitted key findings

Score: 1/6 (Fidelity: 0, Neutralisation: 1, Clarity: 0)

Example 2: Perfect Performance (Qwen, Turner Syndrome)

Source: Primary amenorrhoea in phenotypically appropriate patient

Output: Clean narrative explaining Turner syndrome (45,X) without option references, preserving all clinical facts

Score: 6/6 (Fidelity: 2, Neutralisation: 2, Clarity: 2)

Example 3: Style Over Substance (Multiple models, Anti-CoT Prompt)

Pattern: Grammatically perfect, fluent clinical narratives with fabricated content

Example: Lithium toxicity case generated "hypokalemia, hyperkalemia, cardiac arrhythmias, intravenous lithium chelation therapy" (none present in source)

Lesson: Fluency metrics (BLEU, perplexity) insufficient for medical text evaluation